

Circles: Parts, Circumference, and Area (with π)

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Date: 2026-05-28

Standard: CCSS.MATH.CONTENT.7.G.B.4 — circumference and area of a circle. Key formulas: $\pi = \frac{C}{D}$, $C = \pi D = 2\pi r$, $A = \pi r^2$.

Part A · Parts of a Circle

Problem 1

The segment from the center to the edge.

vocabulary

radius

Problem 2

Chord through the center; how many times the radius?

vocabulary

$D = 2r$

The diameter goes across the whole circle through the center; it equals two radii.

diameter | 2 times

Problem 3

Match the labels a–e on the diagram.

vocabulary

a = center (the dot in the middle)

b = radius (center to edge)

c = diameter (edge to edge through the middle)

d = chord (edge to edge, but NOT through the middle)

e = arc (a curved piece of the edge)

a center, b radius, c diameter, d chord, e arc

Problem 4

Distance around vs. space inside.

vocabulary

Around the outside = circumference. Flat space inside = area.

circumference | area

Part B • Radius ↔ Diameter

Problem 5

$r = 6$ cm. Find D .

$D = 2r$

$D = 2r = 2 \times 6 = 12$

$D = 12$ cm

Problem 6

$D = 18$ in. Find r .

$r = D/2$

$r = \frac{D}{2} = \frac{18}{2} = 9$

$r = 9$ in

Part C · Circumference ($\pi = 3.14$)

Problem 7

$r = 5$ m. Find $C = 2\pi r$.

$$C = 2\pi r$$

$$C = 2\pi r = 2 \times 3.14 \times 5$$

$$C = 6.28 \times 5 = 31.4$$

$$C = 31.4 \text{ m}$$

Problem 8

$D = 10$ cm. Find $C = \pi D$.

$$C = \pi D$$

$$C = \pi D = 3.14 \times 10 = 31.4$$

$$C = 31.4 \text{ cm}$$

Problem 9

Pizza, $D = 14$ in. Find the circumference.

$$C = \pi D$$

word problem

$$C = \pi D = 3.14 \times 14 = 43.96$$

$$C = 43.96 \text{ in}$$

Part D · Area ($A = \pi r^2$)

Problem 10

$r = 4$ cm. Find A , $\pi = 3.14$.

$$A = \pi r^2$$

Square first: $r^2 = 4^2 = 16$

$$A = \pi r^2 = 3.14 \times 16 = 50.24$$

$$A = 50.24 \text{ cm}^2$$

Problem 11

$D = 20$ ft. Find r then A , $\pi = 3.14$.

$$r = D/2$$

$$A = \pi r^2$$

$$r = \frac{20}{2} = 10$$

$$r^2 = 10^2 = 100$$

$$A = 3.14 \times 100 = 314$$

$$A = 314 \text{ ft}^2$$

Problem 12

$r = 7$ cm. Find A using $\pi = \frac{22}{7}$.

$$A = \pi r^2$$

$$\pi = 22/7$$

$$A = \frac{22}{7} \times 7^2 = \frac{22}{7} \times 49$$

$$= 22 \times \frac{49}{7} = 22 \times 7 = 154 \text{ (the 7 cancels!)}$$

$$A = 154 \text{ cm}^2$$

Problem 13

$D = 28$ m. Find r then A using $\pi = \frac{22}{7}$.

$$r = D/2$$

$$\pi = 22/7$$

$$r = \frac{28}{2} = 14$$

$$A = \frac{22}{7} \times 14^2 = \frac{22}{7} \times 196 = 22 \times 28 = 616$$

$$A = 616 \text{ m}^2$$

Part E · Half & Quarter Circles

Problem 14

Half circle, $r = 6$ cm. Find the area, $\pi = 3.14$.

$$A = \pi r^2$$

$$\text{half} = \frac{1}{2}$$

Full circle: $A = 3.14 \times 6^2 = 3.14 \times 36 = 113.04$

Half of it: $\frac{113.04}{2} = 56.52$

$$A = 56.52 \text{ cm}^2$$

Problem 15

Quarter circle, $r = 10$ in. Find the area, $\pi = 3.14$.

$$A = \pi r^2$$

$$\text{quarter} = \frac{1}{4}$$

Full circle: $A = 3.14 \times 10^2 = 314$

Quarter of it: $\frac{314}{4} = 78.5$

$$A = 78.5 \text{ in}^2$$

Part F · Explain in Words

Problem 16

Garden circumference $C = 6.28$ m. (a) explain & find r ; (b) find area; (c) bonus: double the radius.

$$C = 2\pi r$$

$$A = \pi r^2$$

reasoning

(a) Since $C = 2\pi r$, I can undo it: divide the circumference by 2π to get the radius.

$$r = \frac{C}{2\pi} = \frac{6.28}{2 \times 3.14} = \frac{6.28}{6.28} = 1. \text{ So } r = 1 \text{ m.}$$

$$(b) A = \pi r^2 = 3.14 \times 1^2 = 3.14 \text{ m}^2.$$

(c) **In words first:** Circumference grows in a "fair" way — it is a length, so doubling the radius simply **doubles** the distance around. Area is different because it fills a 2-D space (length $\times$ width feel), so doubling the radius makes it **four times** as big.

Check: New $r = 2$. New $C = 2\pi(2) = 4\pi = 12.56$ m $\rightarrow$ that is $2 \times$ the old 6.28. $\checkmark$ New

$A = \pi(2)^2 = 4\pi = 12.56 \text{ m}^2 \rightarrow$ that is $4 \times$ the old 3.14. $\checkmark$

$$r = 1 \text{ m, } A = 3.14 \text{ m}^2; \text{ double } r \rightarrow C \times 2, A \times 4$$

